Pan-Cancer EMT State Atlas

- Clinician-facing research summary. Not a treatment recommendation.

- Current build: EMT states, confidence, clinical screens, single-cell validation, spatial pathology queue, and target hypotheses.

- New validation milestone: first real 3CA annotated dataset imported for Head and Neck Cancer.

- Use boundary: all findings require independent validation and pathology review before clinical interpretation.

Disease-Group EMT Summary

cancer_group	sample_count	mesenchymal_like_fraction	stromal_dominant_fraction	high_confidence_fraction
Skin Cancer	553	0.591	0.0723	0.519
Head and Neck Cancer	566	0.498	0.26	0.3
Blood Tumors	221	0.416	0.0226	0.394
Pancreatic Cancer	183	0.328	0.322	0.0929
Other / Not Requested	4.78e+03	0.256	0.118	0.4
Oesophageal Cancer	196	0.25	0.163	0.48
Brain Tumors	701	0.193	0.0171	0.175
Lung Cancer	1.13e+03	0.156	0.129	0.328
Breast Cancer	1.22e+03	0.129	0.12	0.198
Ovarian Cancer	309	0.0324	0.0259	0.57
Colon / Intestine Cancer	662	0.0287	0.0272	0.606
Prostate Cancer	550	0.0127	0.0127	0.245

Clinical Readiness

cancer_group	sample_count	survival_events	stage_available_fraction	treatment_outcome_available_fraction	stromal_warning_fraction
Other / Not Requested	4.78e+03	1.26e+03	0.661	0.507	0.194
Breast Cancer	1.22e+03	198	0.995	0	0.643
Lung Cancer	1.13e+03	461	0.999	0.82	0.328
Brain Tumors	701	271	0	0.712	0.0171
Colon / Intestine Cancer	662	140	0.979	0.571	0.205
Head and Neck Cancer	566	253	0.873	0.82	0.284
Skin Cancer	553	240	0.928	0	0.0723
Prostate Cancer	550	10	0	0.938	0.0782
Ovarian Cancer	309	185	0	0.812	0.282
Blood Tumors	221	123	0	0.217	0.0226
Oesophageal Cancer	196	82	0.903	0.74	0.24
Pancreatic Cancer	183	95	0.995	0.967	0.858

Annotated Single-Cell Tumor vs Stroma

source	dataset	cancer_grou	cells	malignant	caf_	stroma	canc	myeloid	intrinsic	caf_stroma	normal	myeloid	malignant	caf_stroma	canc	myeloid	malignant	vs_stroma	interpretati
3CA	Puram201	Head and	5.9e+03	2.54e+03	1.44e+03	149	153	1.01e+03	75	-0.584	0.761	0.482	-1.34	caf_stroma
CancerSE	EXP0047	Blood Tum	1.34e+03	1.34e+03	0	0	335	0	0	0				malignant_a
CancerSE	EXP0057	Brain Tum	991	991	0	0	248	0	0	0				malignant_a
CancerSE	EXP0053;	Breast Ca	440	440	0	0	112	0	0	1.33e-16				malignant_a
CancerSE	EXP0051	Colon / In	293	293	0	0	74	0	0	1.52e-18				malignant_a
CancerSE	EXP0063	Head and	2.22e+03	2.22e+03	0	0	554	0	0	3.21e-17				malignant_a
CancerSE	EXP0068	Lung Can	3.52e+03	3.52e+03	0	0	881	0	0	8.07e-18				malignant_a
CancerSE	EXP0023	Oesophag	314	314	0	0	79	0	0	-1.06e-16				malignant_a
CancerSE	EXP0069	Ovarian C	244	244	0	0	61	0	0	1.46e-17				malignant_a
CancerSE	EXP0037	Pancreatic	96	96	0	0	24	0	0	4.63e-17				malignant_a
CancerSE	EXP0070	Prostate C	122	122	0	0	31	0	0	-2.73e-18				malignant_a
CancerSE	EXP0071	Skin Canc	1.26e+03	1.26e+03	0	0	315	0	0	-5.65e-17				malignant_a

Top Disease Target Hypotheses with Lineage Evidence

cancer_group	target	family	disease_priority_score	cell_lines	lineage_evidence_note
Blood Tumors	AXL	AXL_TAM	0.471	142	lineage_dependency_limite
Blood Tumors	MERTK	AXL_TAM	0.403	142	lineage_dependency_limite
Blood Tumors	TYRO3	AXL_TAM	0.374	142	lineage_dependency_limite
Blood Tumors	GAS6	AXL_TAM	0.363	142	lineage_dependency_limite
Blood Tumors	ITGAV	FAK_SRC_integrin	0.312	142	lineage_dependency_limite
Blood Tumors	PTK2	FAK_SRC_integrin	0.311	142	lineage_dependency_limite
Blood Tumors	ITGB1	FAK_SRC_integrin	0.31	142	lineage_dependency_limite
Blood Tumors	SRC	FAK_SRC_integrin	0.31	142	lineage_dependency_limite
Blood Tumors	PROS1	AXL_TAM	0.31	142	lineage_dependency_limite
Blood Tumors	TGFBR1	TGF_beta_SMAD	0.293	142	lineage_dependency_limite
Blood Tumors	SLC2A1	hypoxia_metabolism	0.29	142	lineage_dependency_suppo
Blood Tumors	PXN	FAK_SRC_integrin	0.289	142	lineage_dependency_limite
Brain Tumors	AXL	AXL_TAM	0.412	91	lineage_dependency_limite
Brain Tumors	MERTK	AXL_TAM	0.353	91	lineage_dependency_limite
Brain Tumors	TYRO3	AXL_TAM	0.328	91	lineage_dependency_limite
Brain Tumors	GAS6	AXL_TAM	0.318	91	lineage_dependency_limite
Brain Tumors	ITGAV	FAK_SRC_integrin	0.273	91	lineage_dependency_suppo
Brain Tumors	PTK2	FAK_SRC_integrin	0.272	91	lineage_dependency_suppo
Brain Tumors	ITGB1	FAK_SRC_integrin	0.272	91	lineage_dependency_limite
Brain Tumors	SRC	FAK_SRC_integrin	0.272	91	lineage_dependency_limite
Brain Tumors	PROS1	AXL_TAM	0.271	91	lineage_dependency_limite
Brain Tumors	TGFBR1	TGF_beta_SMAD	0.257	91	lineage_dependency_limite
Brain Tumors	SLC2A1	hypoxia_metabolism	0.254	91	lineage_dependency_limite
Brain Tumors	PXN	FAK_SRC_integrin	0.253	91	lineage_dependency_limite

Spatial Pathology Review Queue

run	cancer_group	pathology_priority_score	review_focus	minimum_review_regions
10x_prostate_ffpe	Prostate Cancer	0.17	tumor_intrinsic_emt;tam_neig	10
geo_pdac_gse274103	Pancreatic Cancer	0.169	tumor_intrinsic_emt;hypoxia_	10
geo_brain_met_gse179572	Brain Tumors	0.169	tumor_intrinsic_emt;tam_neig	10
10x_lung_lusc_cytassist	Lung Cancer	0.169	tumor_intrinsic_emt;caf_stron	10
geo_met_pdac_gse274557_pt	Pancreatic Cancer	0.168	tumor_intrinsic_emt;caf_stron	10
10x_colorectal_intestinal_ffpe	Colon / Intestine Cancer	0.168	caf_stromal_interface;hypoxia	10
10x_breast_ffpe_idc	Breast Cancer	0.167	tumor_intrinsic_emt;caf_stron	10
10x_ovarian_ffpe	Ovarian Cancer	0.166	caf_stromal_interface;invasiv	10